

Combinatorics

Today's Topic

Sum Rule

Product Rule

Product Rule

TASK T

n_1 ways to do task T_1 ,
 n_2 ways to do task T_2



T_1 T_2

* for each of n_1 ways
we can do n_2

Total ways = $n_1 \times n_2$

* both the tasks are
happening
(simultaneously)

Example :-

$\textcircled{T} \left\{ \begin{array}{l} K=0 \\ \text{for } i=1 \text{ to } n_1 (T_1) \leftarrow \\ \text{for } i=1 \text{ to } n_2 (T_2) \leftarrow \end{array} \right. \left\{ \begin{array}{l} i=1, j=1 \text{ to } n_2 \\ i=2, j=1 \text{ to } n_2 \\ \vdots \end{array} \right.$

$$K = K + 1$$

$$T = (n1 * n2)$$

$$i = n1$$

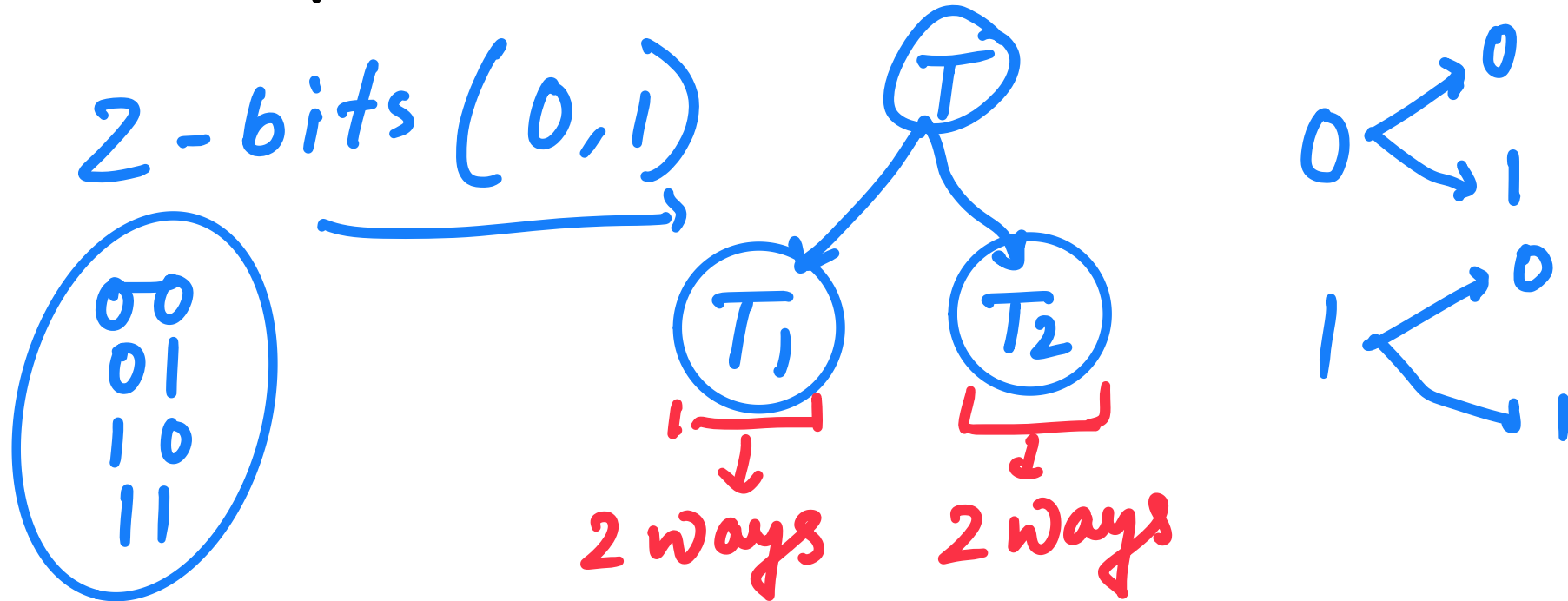
for ($i=1; i \leq n; i++$) $\rightarrow O(n)$.

for ($j=1; j \leq n; j++$) $\rightarrow O(n)$

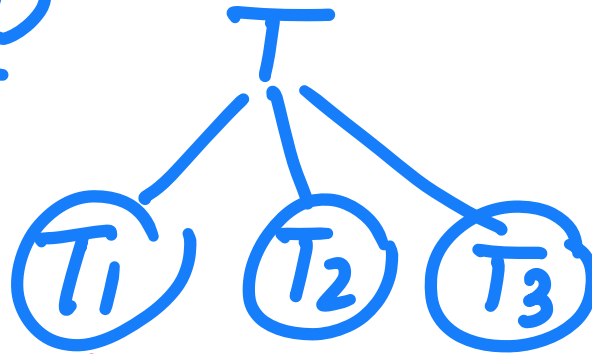
$$O(n) * O(n) \\ = O(n^2).$$



→ How many codes we can generate if we have 2 bits? (n -bits)



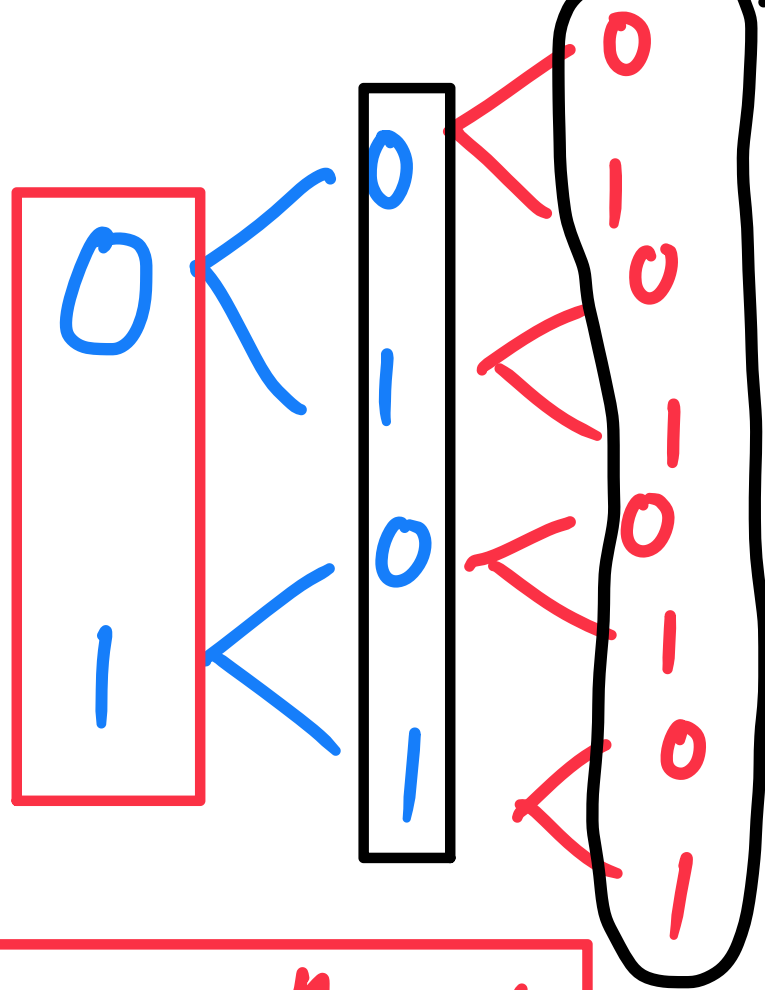
3-bit (0,1)



↓ ↓ ↓
2 ways 2 ways 2 ways

∴ $(2 * 2 * 2) = 8 \text{ ways}$

Tree diagram

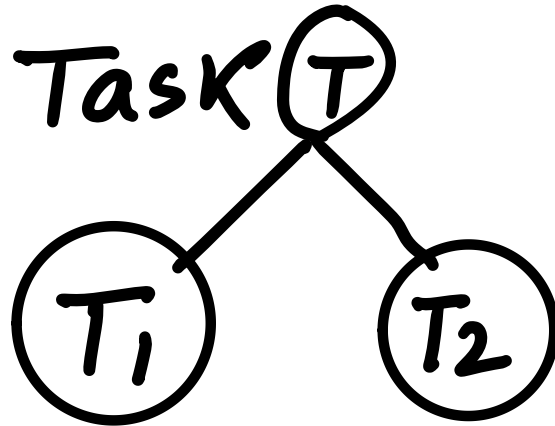


$\therefore n\text{-bits} \Rightarrow 2^n \text{ bits}$

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$\rightarrow 2^3$
 $= 8 \text{ bit}$

Sum Rule:-

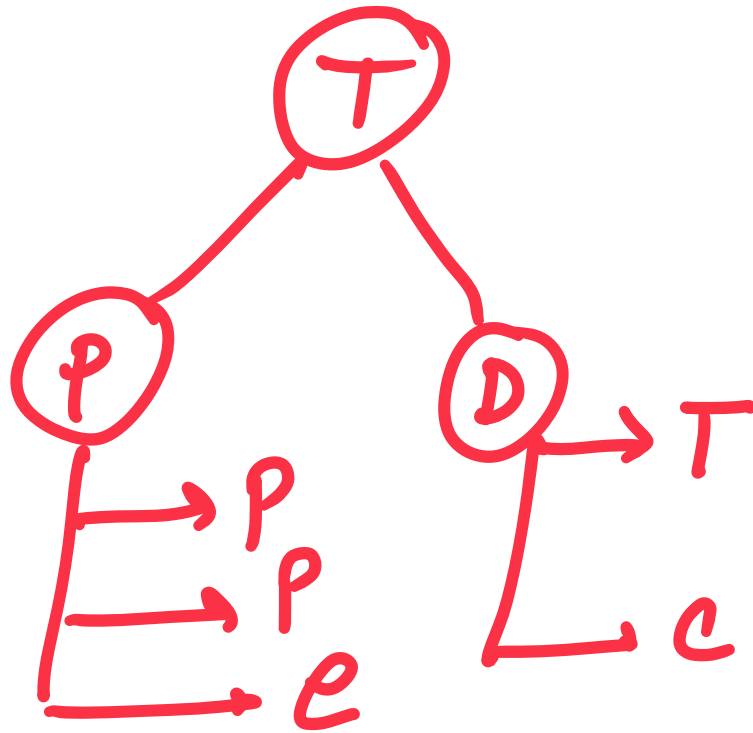


n_1 ways to do task T_1
 n_2 " " " " T_2
 both the task are
 not happening
 simultaneously

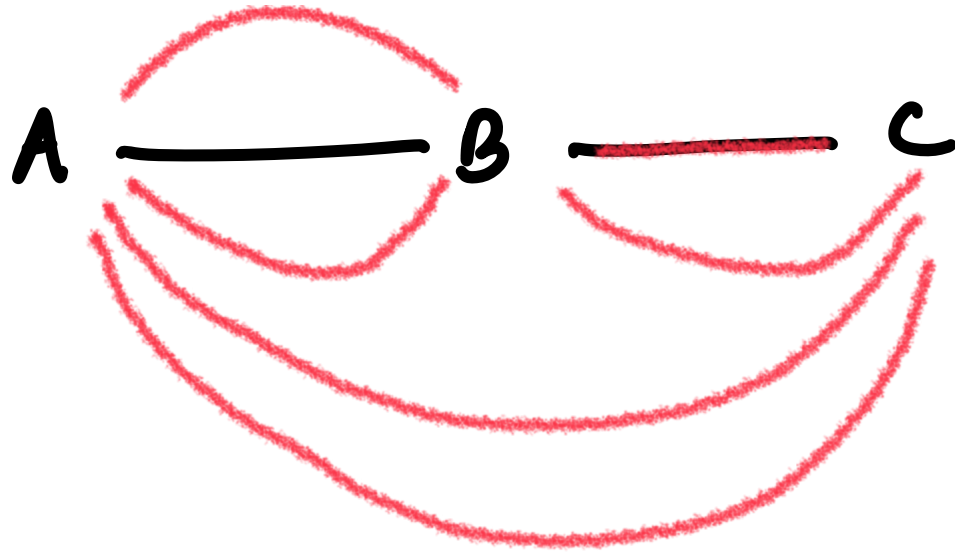
$$\therefore \text{Total ways} = n_1 + n_2$$

lady, she wants to go for shopping
if she does shopping in Patna, she
can buy pen, pencil, eraser. if she
does shopping in delhi, she will
buy table, chair.
How many ways she can do the shopping?





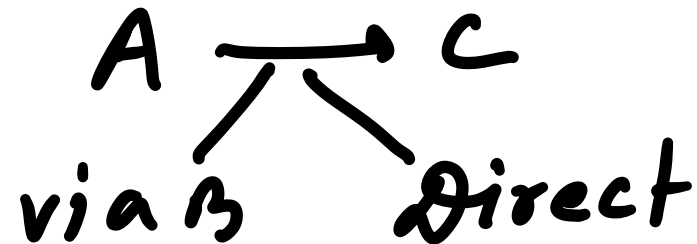
$\therefore$ Total ways
 $= 3 + 2$
 $= 5 \text{ ways}$



1) how many ways we
can go from A to C
(via-B)

3 x 2 ways
= 6 ways.

3) A - C - A (via-B) 2)

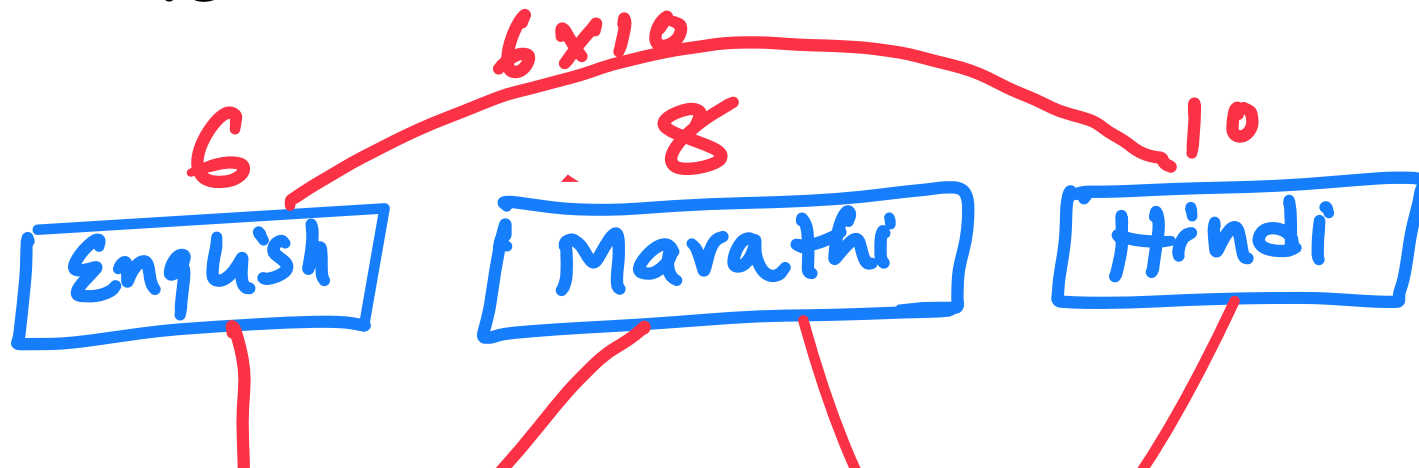


3 x 2 + 2 ways





Suppose we have 6 different English Movies, 8 diff. Marathi Mo., 10 diff. Hindi Movies. How many ways we can choose 2 movies of diff. lang.?



6×8

8×10

$$= 6 \times 8 + 8 \times 10 + 6 \times 10 \text{ ways}$$
$$= \rightarrow ?$$

* A store carries eight styles of pants. For each style, there are 12 different possible waist sizes, five pants lengths and four colour choices. How many different types of pants could the store have?

$$\rightarrow 8 \times 12 \times 5 \times 4.$$

10 $\rightarrow$ English book

6 $\rightarrow$ French "

4 $\rightarrow$ German "

} Given

Q. How many ways are there to select one book?

$$10 + 6 + 4 \rightarrow \text{Sum Rule}$$

Q. How many ways are there to select three books, one of each lang?
 $10 \times 6 \times 4 \rightarrow$ Product Rule

* There are 6 men and 5 women in a room. In how many ways we can choose 3 men and 2 women from Room?
Simultaneously

$$\begin{array}{l} 6M \rightarrow 3M \\ 5W.M. \rightarrow 2W.M. \\ = {}^6C_3 \quad ? \quad {}^5C_2 \end{array}$$

$${}^6C_3 \times {}^5C_2 = 20 \times 10 = 200 \text{ ways.}$$

Q. The no. of 4 digit even numbers that can be formed using 0, 1, 2, 3, 4, 5, 6 without repetition is —

(A) 120

(B) 300

4-digit :- 0, 2, 4, 6

Case-1 $\underline{4} \times \underline{5} \times \underline{6} \times \underline{0} = 120$

- (C) 420
 (D) More than one of the
 above
 (E) None

or
Case-11

$$\underline{5} \times \underline{5} \times \underline{4} \times \underline{3 \text{ ways}} = 300$$

(2 or 4 or 6)

$$(120 + 300) = 420 \text{ Nos}$$

* Three man have 6 coats, 5 belts and 4 caps. The no. of ways they can wear them is _____?

☒ (A) 172800

(B) 172500

(C) 174800

(D) None of the above

Total ways =

$$\begin{aligned}
 & (\text{Coat}) \times (\text{Belt}) \times (\text{Caps}) \\
 & = (6 \times 5 \times 4) \times (5 \times 4 \times 3) \times (4 \times 3 \times 2) \\
 & \quad \quad \quad M_1 \quad M_2 \quad M_3 \quad \quad \quad M_1 \quad M_2 \quad M_3 \quad \quad \quad M_1 \quad M_2 \quad M_3 \\
 & = 172800.
 \end{aligned}$$

Designed and written by Prof. Sanjay Kr. Sen
 Quad A1, School of Tech.
 Dept. of computer science
 (AI/ML)